Reachability · Creativity

# Paid on delivery. Proven on the ledger.

ResearchAgent gives coding agents a research wallet.
Ask a question. Set a budget. Check the delivery key.

collaboration95.github.io/theFastandtheFungible/try/

Open the demo or verify a recorded run.

Watch a recorded run



# From a prototype to a verifiable research wallet

### At SingHacks

One web page. Synthetic sources. A person approved each quote; simulated payments were the default.

### Today

Real documentation as sources. Payments tied to the article key, re-read from a public node for every one of 1,000 purchases. Agents connect by URL.

**339**
merged pull requests since 21 Sep

**15**
published releases

**61**
recorded ledger transactions

**224**
shared verifier cases

# Real sources, and reports signed to a ledger

### Free tier: the real docs

12,104 passages from the XRPL and x402 documentation, each cited by its real page URL, with licences and upstream versions pinned.

### Paid tier: ledger analytics

Reports on RLUSD supply and holders, the RLUSD/XRP pool and project accounts, each signed “as of validated ledger N”. Check one on the public explorer.

### No fictional writers

The real-content profile leaves out all eight sample publishers. Writer provenance sits on each writer’s badge.

The agent cites a passage by its original page, or by the ledger it was computed from.

# Open to anyone, with a regulated mode

### From a phone

Write a note, pass moderation and go live in 12.4 s. A verified sale followed after 34.5 s, measured locally.

### Moderated before it sells

Prechecks catch secrets, seed phrases, contact details and spam, and an operator approves each note.

### Regulated mode, available as a mode

Credentialed writers only, a pinned OFAC XRP address screen, and every zero-charge refusal recorded in an audited log. Optional, and off on the hosted demo.

Policy budgets, hash-chained audit history and offline receipt checks make the spending path inspectable.

— Technical depth

# One question, one budget, and only code can spend it

1
### Ask
Set a question and budget.

2
### Confirm
A five-second plan sheet.

3
### Search
Free abstracts and signed promises.

4
### Decide
Policy rules decide purchases.

5
### Pay
Lock payment in native escrow.

6
### Verify
Check delivery and cite passages.

## Five hard gates

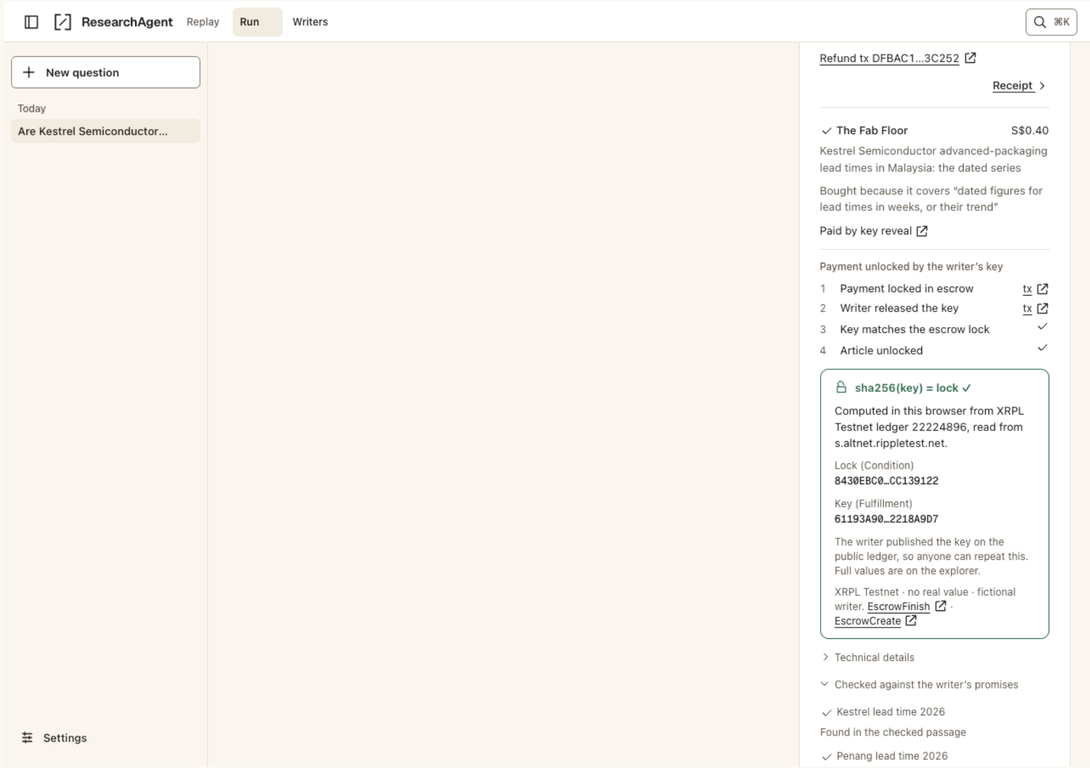
- No premium bytes before a grant.
- Only policy code spends, within the budget.
- One charge per intent, across retries and restarts.
- Real citations to delivered passages.
- Simulated or substituted work is labelled.

# The article key releases the writer’s payment

- The writer encrypts the article and signs its promises.
- Policy locks payment to the article’s key hash.
- The writer publishes the key to collect payment.
- The app grants access after ledger validation.

## Your browser checks the published key against the escrow lock.

Opt-in holdback: 20% of the price waits in a second escrow. Your policy releases it when the promise checks pass, or takes it back after a deadline, with no writer cooperation. 17 real purchases: 11 released, 6 refunded.




Local replay of a recorded escrow · fictional writer · public-node key check, 11 Nov.

— Technical depth

# Proof at scale

## 1,000

**real purchases, each re-read from a public node**

An independent script checked every transaction. Median 20.3 s end to end. Anyone can repeat the check with one command.

## 210,480

**protocol states checked**

No violation, and all seven deliberately seeded bugs caught.

## 1,000,000

**simulated crash and retry schedules**

Run against the real purchase code: no violation.

## 34

**attacks on the local stack**

No breach.

Paid notes help: on 100 XRPL and x402 questions, correct answers rose from 31 with free docs alone to 48 with paid notes.

— Reachability

# Connect your agent by URL

### MCP over HTTP

Paste one URL. One-click add for Cursor, VS Code and Claude; Claude Code takes one command.

### Sell over x402

Express, Hono and FastAPI packages add key-reveal delivery to any paid endpoint. A separate Python buyer completed a real purchase.

### Sell with a package

Express, Hono and FastAPI packages add key-reveal delivery. A separate Python buyer completed a purchase and an expiry cancellation against one.

### SDKs and verifiers

TypeScript and Python can ask, stream and verify, with separately written receipt checkers.

### Agent hires agent

ResearchAgent researches for a caller and pays cited-writer royalties.

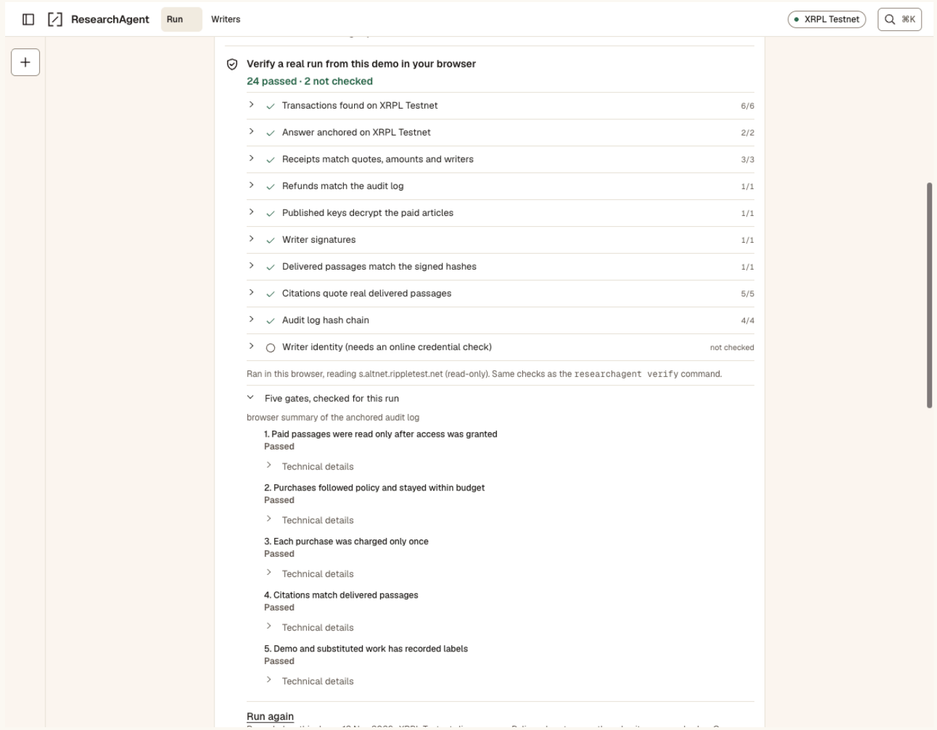
### An x402 scheme

A key-reveal escrow scheme for x402 is drafted for upstream.

```
npx -y https://github.com/Collaboration95/theFastandtheFungible/releases/download/v1.1.1/researchagent-mcp-1.1.1.tgz try --testnet
```

Or run it all from one package command. Recorded release run, 11 Nov: fixture models and sample writers; terminal output is preserved in the evidence file. The calling model cannot raise budget caps.

# One hosted run. All five gates held.




- **Browser / published TypeScript:** 24 passed, 0 failed, two not checked.
- **Published Python:** 25 passed, 0 failed, two skipped.
- Paid passages were read after access was granted.
- One article for S$0.40 in 31.1 s Start→Done, excluding planning and the modal.

Recorded hosted run · 12 Nov.

This run used live DeepSeek planning/final writing, an initial draft from recorded extracts and keyword search. Policy rules decided purchases.

— Feasibility

# Live usage, measured cost, auditable purchases

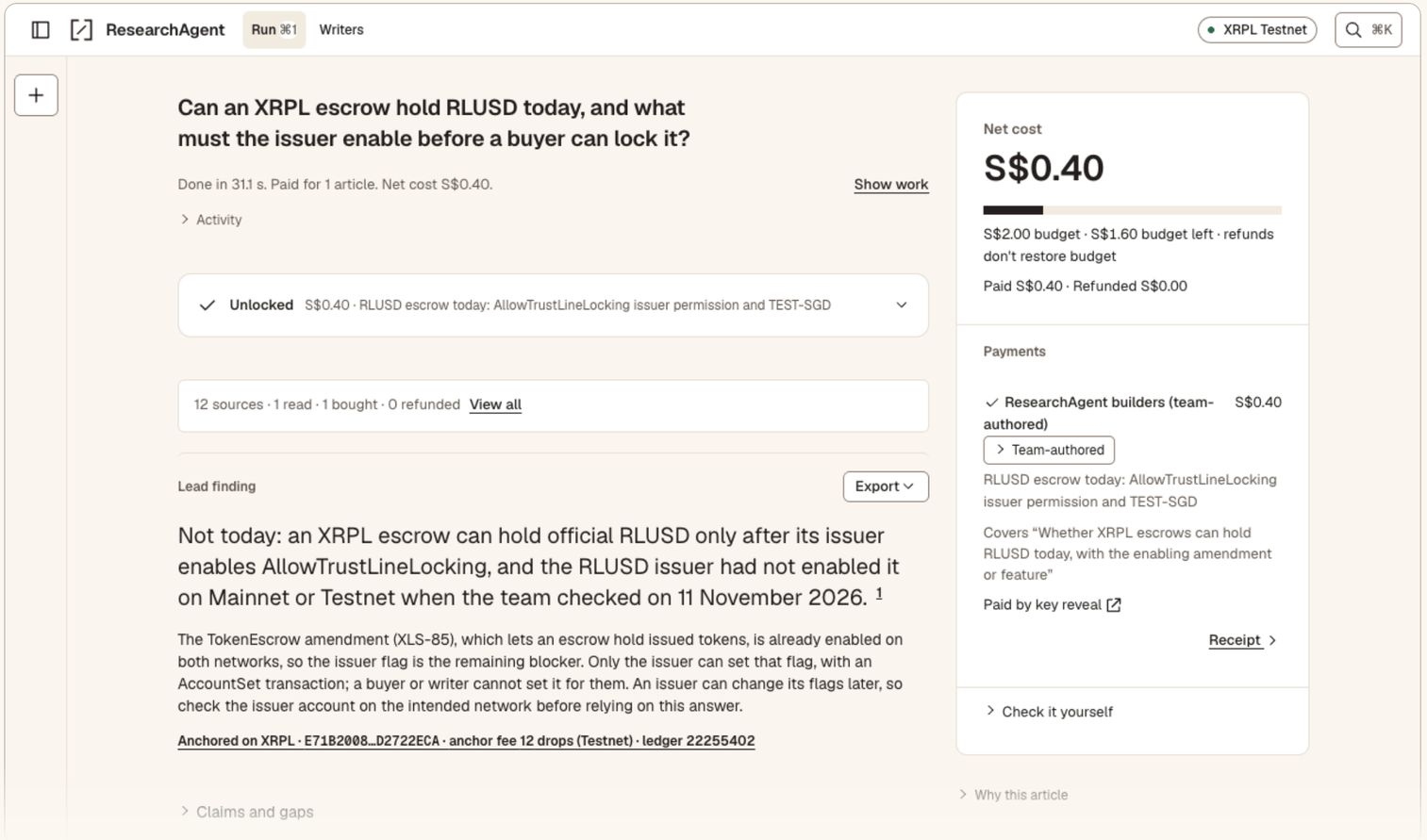
- Hosted XRPL Testnet demo since 11 Nov, team test runs included, as of 12 Nov 2026, 12:24 SGT

| | | |
|---|---|---|
| **62**<br>questions | **74**<br>settled purchases | **US$0.0020035**<br>median model cost per question |
| **22.316 s**<br>median answer time | **61.061 s**<br>p90 · 49 timed runs | **0 violations**<br>across 50 audited runs; 12 audits unavailable |

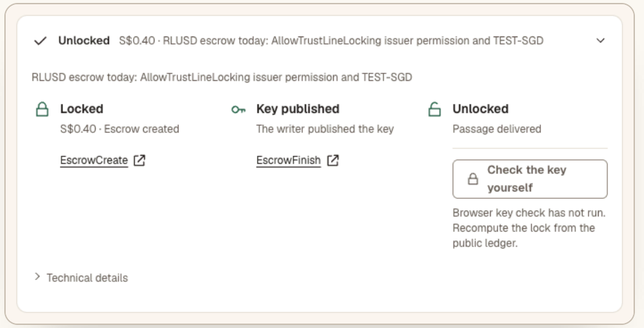
Model cost is separate from article prices and ledger fees. Available audit traces define the violation population.

The live page lists every validated escrow and key check, and the scoreboard separates outside visitors from operator tests.

# Answer, payment, delivery: one view




Hosted answer view, captured 12 Nov.




The payment steps, beside the answer.

# Who buys first

| Buyer and task | Seller | Benefit |
| --- | --- | --- |
| XRPL/x402 integrators using coding agents to implement a paid endpoint or diagnose failed settlement. | Ledger-analytics reports and notes from two team-authored writers, over free documentation. Anyone can publish through moderation. | Buy the specific implementation detail a free abstract omits, with a cited passage and checkable payment. |

Example: how writers get paid.

A research wallet where coding agents already work.

— Builder feedback

# RLUSD cannot be escrowed today

### Issuer locking is off

Ripple’s official RLUSD issuers have escrow locking switched off on both networks. Unsigned probes return `tecNO_PERMISSION`; the matched stand-in token succeeds. We show the verdict before anyone signs.

### Batch success can mislead

A Batch makes two escrows atomic, but the outer transaction reports success even when its inner transactions roll back.

### Delegate one permission

A visitor can let a service create escrows and nothing else, then revoke it.

Five upstream items with transaction hashes and three patches are ready to file. 7 ranked findings with runnable repros · 35 documented issues · 15 submissions on 4–5 Sep · 8 earlier upstream drafts, 0 filed.

Guruprasath · product and engineering
Miul · research-plan contracts and test harness
Andrew Nguyen · initial idea briefs and data sources

— Feasibility · Reachability

# Try it. Check the key.

## Status and limits

- Development network only; mainnet signing disabled. No outside customers, pilots or revenue; the 1,000 purchases are scripted repeats of one team-authored note. Demand is not yet validated.
- Signed submission can expose the key before validation; escrow stays claimable until the escrow deadline. Key reveal proves delivery, not quality.
- Only the holdback share is refundable without the writer; the key-release payment stays paid. No-show recovery needs ledger-confirmed escrow cancellation after the deadline.
- Records include accepted operator-issued Credentials and an operator-issued Permissioned Domain. Operator identity is not KYC, quality or payment permission. Shared payer and visitor caps; cold installs depend on npm and faucet latency.
- Registry publication, Mac signing, regulated custody, business identity checks and licensing work remain; the OFAC screen is one dated address list, not a clearance. Run-specific substitutions are labelled with their evidence.

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Limits and prepared answers: docs/submission/qa-prep.md